

# LLM-Based Assistant for the OpenROAD Project

## OpenROAD AI Assistant

An LLM-Based Assistant for the OpenROAD Chip Design Flow

### 1. Introduction

OpenROAD is an open-source Electronic Design Automation (EDA) tool designed to automate the chip design process from RTL to GDSII. Although the platform provides powerful functionality, it primarily operates through a command-line interface and requires users to understand complex commands, scripts, and documentation.

New users often face difficulties such as:

- Understanding OpenROAD commands
- Navigating large documentation
- Writing correct Tcl scripts
- Debugging errors in log files

To address these challenges, this project proposes an AI-powered assistant for OpenROAD that enables users to interact with the system using natural language queries.

### 2. Project Objective

The objective of this project is to develop an LLM-powered AI assistant that simplifies the usage of OpenROAD by providing intelligent support for:

- Command explanation
- Script generation
- Error debugging
- Documentation search
- Design flow guidance

The assistant is built using a Retrieval-Augmented Generation (RAG) architecture combined with a local Large Language Model.

### 3. System Features

#### Command Explanation

The system can explain OpenROAD commands in simple terms.

**Example:**

**User query:**

***“How do I run global placement in OpenROAD?”***

The assistant explains the `global_placement` command and its parameters.

## **Script Generation**

The assistant can automatically generate Tcl scripts for different stages of the chip design flow, including:

- Floorplanning
- Placement
- Clock Tree Synthesis
- Routing

## **Error Debugging**

Users can paste OpenROAD error logs, and the system will:

- Identify the root cause
- Provide explanations
- Suggest possible solutions

## **Documentation Search**

The system provides semantic search across OpenROAD documentation and returns relevant information quickly.

## **Design Flow Guidance**

The assistant can guide users step-by-step through the RTL-to-GDSII design process.

## **4. System Architecture**

The system is divided into two major components:

### **Frontend**

- Web interface
- User interaction
- Query submission

### **Backend**

- RAG retrieval engine
- LLM processing
- Documentation search
- API services

## **5. RAG (Retrieval-Augmented Generation) Pipeline**

The assistant follows the RAG architecture to ensure reliable responses.

### **Process:**

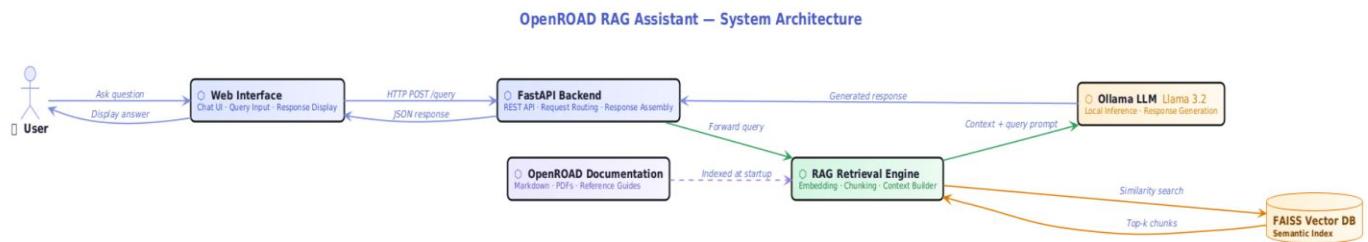
1. User submits a query
2. The system converts the query into embeddings
3. FAISS vector search retrieves relevant documentation

4. The retrieved context is sent to the LLM
5. The LLM generates a response
6. The response is returned to the user

## 6. Technologies Used

| Component       | Technology              |
|-----------------|-------------------------|
| Backend         | Python                  |
| API Framework   | FastAPI                 |
| LLM             | Llama 3.2 (via Ollama)  |
| Vector Database | FAISS                   |
| Embedding Model | Sentence Transformers   |
| Frontend        | HTML / CSS / JavaScript |

## 7. System Architecture Diagram (PlantUML)



## 8. Expected Outcome

The OpenROAD AI Assistant will:

- Improve accessibility of OpenROAD for beginners
- Reduce time required to understand commands
- Simplify debugging of OpenROAD errors
- Provide intelligent guidance during chip design workflows

This system demonstrates how Artificial Intelligence can enhance usability in Electronic Design Automation tools.